

A Novel Active Contour Model for Noisy Image Segmentation Based on Adaptive Fractional Order Differentiation

Meng-Meng Li and Bing-Zhao Li^{ID}, *Member, IEEE*

Abstract—The images used in various practices are often disturbed by noise, such as Gaussian noise, speckled noise, and salt and pepper noise. Images with noise are one of the challenges for segmentation, since the noise may cause inaccurate segmented results. To cope with the effect of noise on images during segmentation, a novel active contour model is proposed in this paper. The newly proposed model consists of fitting term, regularization term and penalty term. The fitting term is designed using a Gaussian kernel function and fractional order differentiation with an adaptively defined fractional order, which applies different orders to different pixels. The regularization term is applied to maintain the smoothness of curves. In order to ensure stable evolution of curves, a penalty term is added into the proposed model. Comparison experiments are conducted to show the effectiveness and efficiency of the proposed model.

Index Terms—Image segmentation, active contour model, fractional order differentiation, level set, variational method.

I. INTRODUCTION

IMAGE segmentation is a basic yet still challenging problem in the fields of image processing and computer vision. It aims to separate an image into several distinct constituents, which can help in analyzing and understanding image information. Active contour models (ACM) [1], [2] are widely applied to image segmentation [3], scene understanding [4] and target tracking [5], etc.

In the ACM scheme, the active contour is induced to move towards the image boundaries, which eventually abstracts the target areas inside the image domain. Typically, the active contour is represented implicitly by a higher dimensional level set [6], which was proposed by Osher and Sethian in [7]. The application of level set makes it easier to formulate the energy functional. Since the research of Osher and Sethian, there have been many models and methods developed based on both the ACM scheme and the level

set function representation. In the process of computing image segmentation models, the variational method [8] has been widely applied to minimize energy functionals. Usually Euler-Lagrange equations are deduced and the level set functions are iteratively updated to obtain the segmentation results.

ACMs are generally classified into two types: the edge-based models [9], [10] and the region-based models [11]–[14]. The edge-based models, using the image gradient as an edge stop function, perform well on images with clear edges but have poor performance on images with noise. The region-based models, using statistical intensity information instead of the image gradient, are more robust to noise and the initial contour placements. Therefore the region-based models are better choice for noisy image segmentation.

Among the ACMs, the Chan-Vese (CV) model or the piecewise constant model [15] proposed by Chan and Vese, which was developed from the Mumford-Shah model [16], is one of the most popular region-based models. It assumes that image regions are piecewise-constant and utilizes global intensity fitting, which is not accurate enough to describe the detailed image information. It may become trapped in a local minimum without constraints on the evolving level set function. Based on the CV model, the local binary fitting (LBF) model proposed by Li et.al in [17] utilizes the local property of the Gaussian kernel function to approximate the original image, which is more accurate than global fitting of the CV model. There are also some models considering the local characteristics to improve the accuracy, such as the local region-based Chan-Vese (LRCV) model [18], the local Chan-Vese (LCV) model [19] and the local image fitting (LIF) model [20], etc. In order to solve the problem of falling into the local minimum, in [21] a penalty term is proposed to constrain the level set function to a signed distance function. This completely avoids the reinitialization step and greatly reduces computational cost. Based on these studies, some measures related to image characteristics are considered and combined into segmentation models. For example, in [22], the authors considered the entropy and proposed an ACM that incorporates the edge entropy into their fitting energy. Additionally, in [14], the Kullback-Leibler divergence is utilized to formulate the fitting term. These works are generally classified as CV models or LBF models. However, they may be sensitive to noise and the initial placements of evolving curves.

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The authors are with School of Mathematics and Statistics, Beijing Institute of Technology, Beijing 102488, China, and also with the Beijing Key Laboratory on MCAACI, Beijing Institute of Technology, Beijing 100081, China (e-mail: li_bingzhao@bit.edu.cn; lmmbit@163.com).

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In order to segment images with noise, many researchers have explored different methods and theories applicable to image denoising [23]–[26], hoping to segment images with noise. During this exploration process, fractional order differentiation [27]–[30] has been shown to have great superiority in cases of noise. This occurs because it can non-linearly preserve the low-frequency information and smooth or enhance the high-frequency information, which depends on values of the fractional order. Models in [29]–[31] utilize fractional order differentiation in segmentation models for images with noise or intensity inhomogeneity. They are very innovative methods and more sufficient than the classical models based on the integral order difference. However, the orders of fractional differentiation in those models are fixed constants for given images, which means that all the pixels of the image are with the same differentiation order. The process of choosing the fractional order may be time-consuming and need repeated experiments to obtain desiring values. A good strategy in [30] is designing an adaptive weighting based on local contrast ratio of image gradients, which reduces the number of tunable parameters in the model. In addition to the ACM based methods, learning-based techniques are also widely explored in some recent studies, such as [32]–[34], etc. Different networks designed for the segmentation task have obtained great performance. However, the learning-based methods require labeled datasets, which are not always available, especially for medical images. In addition, the size of training dataset may be quite large. Although learning-based methods are not the focus of this paper, the problems arising in these techniques should be acknowledged.

Based on these studies above, in this paper, a novel active contour model based on adaptive fractional order differentiation is proposed for noisy image segmentation. The proposed model consists of fitting term, regularization term and penalty term. Among them, the fitting term is formulated based on both the local intensity and fractional order differentiation. These two parts constitute the whole fitting energy. In order to avoid large number of repeated experiments for choosing suitable fractional differentiation order of images, an adaptively defined fractional differentiation order is proposed. It is briefly defined by scaled gradient amplitudes of test images. Different pixels are assigned different fractional orders according to the definition. Thus the adaptability of fractional order is achieved. A popular regularization term is added to control the perimeter of the level set function. In addition, the application of a penalty term avoids the reinitialization step of the initial curve.

The rest of the paper is organized as follows. In Section II, we give a brief review of some representative models and a short analysis of their limitations. In Section III, we describe the details of the proposal of our model and its simplification. Section IV gives the algorithm of the proposed model. Section V discusses the proposed model in different cases and provides numerical experiments to show the effectiveness and efficiency. Finally, we conclude our work in Section VI.

II. BACKGROUND

In this section, we give a brief review of the CV model [15], the LBF model [17], the LRCV model [18], and the fractional order ACMs [29] and [30].

A. The CV Model [15]

Let Ω be the image domain, $I : \Omega \rightarrow R$ be a gray image, and Γ be the active contour inside Ω , which eventually separates the image domain into distinct regions. The active contour Γ is implicitly represented by a zero level set $\Gamma = \{(x, y) \mid \phi(x, y) = 0\}$, where ϕ is a level set function and is given by

$$\phi(x) = \begin{cases} c, & x \in \Gamma_{in} \\ 0, & x \in \Gamma \\ -c, & x \in \Gamma_{out} \end{cases} \quad (1)$$

where Γ_{in} and Γ_{out} are areas inside and outside of Γ , respectively. The CV model assumes that the image is a piecewise constant function. It can be interpreted from the classification prospective, which devices the intensities into two categories and minimizes the distances between intensities and the clustering center of the same categories. The CV model is written as

$$E^{CV} = \lambda_1 \int_{\Gamma_{in}} (I - c_1)^2 dx dy + \lambda_2 \int_{\Gamma_{out}} (I - c_2)^2 dx dy + \nu \int_{\Omega} |\nabla H(\phi)| dx dy \quad (2)$$

where λ_1, λ_2 and ν are weighting parameters, c_1 and c_2 are the average intensities inside and outside the curve Γ respectively, and Ω is the image domain. $H(\cdot)$ is the Heaviside function and its derivative is the Dirac function $\delta(\cdot)$. c_1 and c_2 are computed by

$$c_i = \frac{\int_{\Omega} I(x) H_i(\phi(x)) dx}{\int_{\Omega} H_i(\phi(x)) dx}, \quad i = 1, 2 \quad (3)$$

where $H_1(\phi) = H(\phi)$, $H_2(\phi) = 1 - H(\phi)$. In practice, H and its derivative $\delta(\phi)$ are typically regularized as follows.

$$H_\varepsilon(\phi) = \frac{1}{2} \left[1 + \frac{2}{\pi} \arctan\left(\frac{\phi}{\varepsilon}\right) \right] \quad (4)$$

$$\delta_\varepsilon(\phi) = \frac{1}{\pi} \frac{\varepsilon}{\varepsilon^2 + \phi^2} \quad (5)$$

In the CV model, the average intensities c_1 and c_2 are modeled from a global perspective. However, the global information is not accurate enough to describe the image details and it is insufficient to segment images with noise and intensity inhomogeneity. Additionally it is vulnerable to be trapped in the local minimum and requires a time-consuming reinitialization step [29].

B. The LBF Model [17]

The LBF model utilizes the local intensity clustering property of a Gaussian kernel function K and is formulated as

follows

$$E^{LBF} = \lambda_1 \int_{\Omega} K(y-x)(I(y) - f_1(x))^2 H(\phi) dx dy + \lambda_2 \int_{\Omega} K(y-x)(I(y) - f_2(x))^2 (1-H(\phi)) dx dy \quad (6)$$

where λ_1 and λ_2 are fitting parameters, f_1 and f_2 are written as follows,

$$f_1 = \frac{\int_{\Omega} K(y-x)[I(x)H(\phi)] dx dy}{\int_{\Omega} K(y-x)H(\phi) dx dy} \quad (7)$$

$$f_2 = \frac{\int_{\Omega} K(y-x)[I(x)(1-H(\phi(x)))] dx dy}{\int_{\Omega} K(y-x)[(1-H(\phi(x)))] dx dy}. \quad (8)$$

The overall LBF energy functional is defined by

$$\mathcal{E}^{LBF}(\phi, f_1, f_2) = E^{LBF} + \frac{\mu}{2} \int_{\Omega} (|\nabla \phi| - 1)^2 dx + \gamma \int_{\Omega} |\nabla H(\phi)| dx \quad (9)$$

where μ and γ are parameters of penalty term and regularization term, $f_1(x)$ and $f_2(x)$ are the approximate values of image intensities inside and outside the contour Γ respectively.

The introduced Gaussian kernel function K , which greatly improves the fitting term by changing from global fitting to local fitting, is of vital importance for capturing more detailed image information. The LBF model achieves more accurate segmentation results. Furthermore, the penalty term penalizing the deviation from a signed distance function [21] completely avoids the reinitialization of ϕ . However, it does not perform well on images with noise.

C. The LRCV Model [18]

In [18], a local region-based Chan-Vese (LRCV) model was proposed for image segmentation. This model is designed for images with inhomogeneities by incorporating a Gaussian kernel function g_k to average intensities inside and outside the curve. The LRCV model is formulated as

$$E^{LRCV} = \lambda_1 \int_{\Omega} (I - c_1)^2 H(\phi) dx + \lambda_2 \int_{\Omega} (I - c_2)^2 (1 - H(\phi)) dx \quad (10)$$

where c_1 and c_2 are computed by

$$c_1(x) = \frac{\int_{\Omega} g_k(x-y)(I(y)H(\phi(y))) dy}{\int_{\Omega} g_k(x-y)H(\phi(y)) dy} \quad (11)$$

$$c_2(x) = \frac{\int_{\Omega} g_k(x-y)(I(y)(1-H(\phi(y)))) dy}{\int_{\Omega} g_k(x-y)(1-H(\phi(y))) dy} \quad (12)$$

It is very efficient and effective for images with inhomogeneities. However, it needs an extra step to smooth the curve after each iteration. It still suffers from sensitivity to the initialization [41].

D. Fractional ACM [29]

The fractional ACM in [29] is designed for noisy images in segmentation problem. In this model, the global fitting is

derived from the CV model and the local fitting uses fractional order differentiation. The energy functional is given by

$$E = \lambda \sum_{i=1}^2 \int_{\Omega} |I - c_i|^2 H_i(\phi) dx dy + \beta E^{FF} + \mu \int_{\Omega} \delta(\phi) p(|\nabla G_{\sigma} * I|) |\nabla \phi| dx dy + \frac{1}{2} \int_{\Omega} (|\nabla \phi| - 1)^2 dx dy \quad (13)$$

where

$$E^{FF} = \sum_{i=1}^2 \int_{\Omega} |D_x^v I - d_i|^2 H_i(\phi) dx dy \quad (14)$$

c_i ($i = 1, 2$) are the same as Eq. (3), $H_1(\phi) = H(\phi)$, $H_2(\phi) = 1 - H(\phi)$, $D_x^v I$ is the v -th order fractional differentiation in x direction of image I , and d_1, d_2 are updated by

$$d_1 = \frac{\int_{\Omega} D_x^v I \cdot H(\phi) dx dy}{\int_{\Omega} H(\phi) dx dy} \quad (15)$$

$$d_2 = \frac{\int_{\Omega} D_x^v I \cdot (1 - H(\phi)) dx dy}{\int_{\Omega} (1 - H(\phi)) dx dy} \quad (16)$$

$p(\cdot)$ is the potential function written as

$$p(s) = 1 + \frac{1}{1 + s^2}. \quad (17)$$

The fractional Fourier transform and fractional calculus [36], [38] are widely used to calculate the fractional differentiation. The fractional differentiation of an image is computed as follows

$$D_x^v I = F^{-1} \left((1 - \exp(-j2\pi w_1/m))^v \hat{I}(w_1, w_2) \right) \quad (18)$$

$$D_y^v I = F^{-1} \left((1 - \exp(-j2\pi w_2/m))^v \hat{I}(w_1, w_2) \right) \quad (19)$$

$\hat{I}$ is the Fourier transform of image I , F^{-1} is the inverse Fourier transform.

This fractional active contour model that incorporates fractional order differentiation into the fitting term is effective to segment images. However, the value of fractional orders need to be selected carefully for different images.

E. Fractional ACM [30]

In [30], based on the CV model, the difference of the original image and its fractional gradient amplitude is incorporated into the model as follows

$$E = (1-w) \left(\int_{\Omega} |I - c_1|^2 H(\phi) dx + \int_{\Omega} |I - c_2|^2 (1 - H(\phi)) dx \right) + w \left(\int_{\Omega} |u - m_1|^2 H(\phi) dx + \int_{\Omega} |u - m_2|^2 (1 - H(\phi)) dx \right) + \mu \int_{\Omega} |\nabla H(\phi)| dX \quad (20)$$

where

$$u = I - |\nabla^a I|$$

α is the order of the fractional differentiation, c_1 and c_2 are from CV model, m_1 and m_2 are calculated by

$$m_1 = \frac{\int_{\Omega} G_{\sigma}(x-y)u(y)H(\phi)dy}{\int_{\Omega} G_{\sigma}(x-y)H(\phi)dy} \quad (21)$$

$$m_2 = \frac{\int_{\Omega} G_{\sigma}(x-y)u(y)(1-H(\phi))dy}{\int_{\Omega} G_{\sigma}(x-y)(1-H(\phi))dy} \quad (22)$$

and w is the adaptive weighting for fitting terms in this model. The adaptability of w is convenient for parameter setting. However, the complexity of the model makes it inefficient for computation. Therefore, more efficient techniques need to be explored.

III. THE PROPOSED MODEL

In this section, a new active model is proposed for noisy image segmentation. The proposed model consists of three components: fitting term E^F , regularization term E^R and penalty term E^P , i.e.,

$$E = E^F + E^R + E^P. \quad (23)$$

In the following, the three terms are formulated in details.

A. Fitting Term

The fitting term includes two parts. One is derived from LBF model, another denoted by E^{Fr} is based on fractional order differentiation. The whole fitting term is formulated as

$$\begin{aligned} E^F = & \lambda \left[\int_{\Omega} K_{\sigma}(y-x)(I(y)-f_1(x))^2 H(\phi(y)) dx dy \right. \\ & + \int_{\Omega} K_{\sigma}(y-x)(I(y)-f_2(x))^2 (1-H(\phi(y))) dx dy \left. \right] \\ & + E^{Fr} \end{aligned} \quad (24)$$

where f_1, f_2 are average intensities of I inside and outside ϕ respectively, K_{σ} is a Gaussian kernel function as follows

$$K_{\sigma}(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{|x|^2}{2\sigma^2}\right) \quad (25)$$

σ is a scale parameter, E^{Fr} is the fractional order differentiation based fitting related to $D^{\nu}I$. $D^{\nu}I$ is the fractional differentiation of image I , the definition of ν will be discussed later. $D^{\nu}I$ can be taken as another image containing different information and characteristics but that is closely related to the original image. For any $(x, y) \in \Omega$, it is obvious that $D^{\nu}I(x, y)$ approximated by intensities around (x, y) is more accurate than by intensities far away from (x, y) . In this sense, a Gaussian kernel is also required in E^{Fr} . Therefore E^{Fr} is defined by

$$\begin{aligned} E^{Fr} = & \beta \left[\int_{\Omega} K_{\sigma}(y-x)(D^{\nu}I(y)-d_1(x))^2 H(\phi(y)) dx dy \right. \\ & + \int_{\Omega} K_{\sigma}(y-x)(D^{\nu}I(y)-d_2(x))^2 (1-H(\phi(y))) dx dy \left. \right] \end{aligned} \quad (26)$$

where $\beta > 0$, d_1 and d_2 are average intensities of $D^{\nu}I$ inside and outside ϕ respectively.

Keeping ϕ fixed and minimizing E^F with respect to f_i and d_i ($i = 1, 2$) respectively, we obtain the formulations of f_i and d_i as follows

$$f_i = \frac{\int_{\Omega} K_{\sigma}(y-x)[H_i(\phi)I] dx dy}{\int_{\Omega} K_{\sigma}(y-x)H_i(\phi) dx dy} \quad (27)$$

$$d_i = \frac{\int_{\Omega} K_{\sigma}(y-x)[H_i(\phi)D^{\nu}I] dx dy}{\int_{\Omega} K_{\sigma}(y-x)H_i(\phi) dx dy} \quad (28)$$

where $H_1(\phi) = H(\phi)$, $H_2(\phi) = 1 - H(\phi)$, $H(\phi)$ is from Eq. (4). The gradient flow of E^F is derived as

$$\begin{aligned} \frac{\partial \phi}{\partial t} = & -\lambda \delta(\phi) \left[\int_{\Omega} K_{\sigma}(y-x)(I(y)-f_1(x))^2 dy \right. \\ & - \int_{\Omega} K_{\sigma}(y-x)(I(y)-f_2(x))^2 dy \left. \right] \\ & - \beta \delta(\phi) \left[\int_{\Omega} K_{\sigma}(y-x)(D^{\nu}I(y)-d_1(x))^2 dy \right. \\ & - \int_{\Omega} K_{\sigma}(y-x)(D^{\nu}I(y)-d_2(x))^2 dy \left. \right]. \end{aligned} \quad (29)$$

In order to improve the computational efficiency, motivated by the degraded CV (DCV) in [18], the gradient flow of E^F is rewritten as

$$\begin{aligned} \frac{\partial \phi}{\partial t} = & -2\lambda(f_1 - f_2)\delta(\phi) \int_{\Omega} K_{\sigma} \cdot \left(I - \frac{f_1 + f_2}{2}\right) dy \\ & - 2\beta(d_1 - d_2)\delta(\phi) \int_{\Omega} K_{\sigma} \cdot \left(D^{\nu}I - \frac{d_1 + d_2}{2}\right) dy. \end{aligned} \quad (30)$$

Regarding $f_1 - f_2$ and $d_1 - d_2$ as constants, the simplified gradient flow of E^F can be formulated by

$$\begin{aligned} \frac{\partial \phi}{\partial t} = & -\delta(\phi) \left[\lambda \int_{\Omega} K_{\sigma} \cdot \left(I - \frac{f_1 + f_2}{2}\right) dy \right. \\ & + \beta \int_{\Omega} K_{\sigma} \cdot \left(D^{\nu}I - \frac{d_1 + d_2}{2}\right) dy \left. \right]. \end{aligned} \quad (31)$$

For ν , the order of fractional differentiation, it is related to image frequencies and is usually in the range of (0,1). In the image domain, the low-frequency and high-frequency regions have smaller and larger gradient amplitudes respectively, which correspond to relatively smooth and noisy regions respectively. Furthermore, image regions with different frequencies are supposed to be treated differently. Therefore the fractional order is defined by

$$\nu(x, y) = \frac{|\nabla I(x, y)|}{\max_{(x,y) \in \Omega} \{|\nabla I(x, y)|\}} \quad (32)$$

where

$$|\nabla I(x, y)| = \sqrt{(\nabla_x I(x, y))^2 + (\nabla_y I(x, y))^2} \quad (33)$$

$\nabla_x I(x, y)$ and $\nabla_y I(x, y)$ are gradients of $I(x, y)$ in x and y directions. Although the fractional order ν is defined by a brief formula, it takes different values for different pixels of the image and realizes adaptive selection for fractional orders. This means that we do not need to repeat large amount of experiments to manually select suitable values of ν for different images. ν is smaller on flat or low-frequency areas and larger on edges, texture or high-frequency areas. It is only computed once without iterative update.

B. Penalty Term

The penalty term is applied to avoid the reinitialization of the curve, which is constrained by a signed distance function [10] as follows

$$E^P = \frac{\mu}{2} \int_{\Omega} (|\nabla\phi| - 1)^2 dx dy \quad (34)$$

The gradient flow of E^P is derived by

$$\frac{\partial\phi}{\partial t} = \mu[\Delta\phi - \operatorname{div}\left(\frac{\nabla\phi}{|\nabla\phi|}\right)] \quad (35)$$

where Δ is the Laplace operator, $\operatorname{div}(\cdot)$ is the divergence.

C. Regularization Term

In order to ensure the smoothness of the evolving curve, the regularization term is explored to constrain the perimeter of the curve. One of the most popular regularization is defined by

$$E^R = \gamma \int_{\Omega} g|\nabla H(\phi)| dx dy \quad (36)$$

where $\gamma > 0$, g is an edge indicator function [10] given by

$$g = \frac{1}{1 + |\nabla K_{\sigma} * I|^2} \quad (37)$$

$\nabla K_{\sigma} * I = K_{\sigma} * \nabla I$ can be regarded as the smoothed image gradient. When $|\nabla K_{\sigma} * I|$ is large enough, g is almost zero and E^R reaches its minimum. In other words, the curve stops evolving when it reaches the edges, which is satisfied with the segmentation task.

The gradient flow of E^R is derived by

$$\frac{\partial\phi}{\partial t} = \gamma g \delta(\phi) \cdot \operatorname{div}\left(\frac{\nabla\phi}{|\nabla\phi|}\right). \quad (38)$$

In practice, we regularize $\delta(\cdot)$ and $H(\cdot)$ using Eq. (5) and Eq. (4) respectively, and still denote them as $\delta(\cdot)$ and $H(\cdot)$.

In order to better analyze the proposed model, the following remarks are provided.

Remark 1: Here we briefly analyze the differences of our model.

Different from [29] and [30], the order of the fractional differentiation in our model is defined by the scaled image gradient (seen in Eq. (32)) and is computed once. It avoids repeating experiments to determine an optimal fractional order.

With the Gaussian kernel K_{σ} , E^{Fr} approximates $D^{\nu}I$ locally rather than globally in [29]. Although the Gaussian kernel is applied in [30], the objective of the fitting is different from ours.

Remark 2: The Gaussian kernel K_{σ} is vital for E^{Fr} of our model. In order to show the effects of K_{σ} more clearly, gradient flows of the proposed model with and without K_{σ} in E^{Fr} are summarized below.

Case 1. The gradient flow of the whole model with the Gaussian kernel in E^{Fr} is as follows

$$\begin{aligned} \frac{\partial\phi}{\partial t} = & -\delta(\phi)[\lambda \int_{\Omega} K_{\sigma} \cdot (I - \frac{f_1+f_2}{2}) dy \\ & + \beta \int_{\Omega} K_{\sigma} \cdot (D^{\nu}I - \frac{d_1+d_2}{2}) dy + \gamma g \cdot \operatorname{div}\left(\frac{\nabla\phi}{|\nabla\phi|}\right)] \\ & + \mu[\Delta\phi - \operatorname{div}\left(\frac{\nabla\phi}{|\nabla\phi|}\right)]. \end{aligned} \quad (39)$$

where d_1 and d_2 are from Eq. (28).

Case 2. The gradient flow of the whole model without the Gaussian kernel in E^{Fr} is as follows

$$\begin{aligned} \frac{\partial\phi}{\partial t} = & -\delta(\phi)[\lambda \int_{\Omega} K_{\sigma} \cdot (I - \frac{f_1+f_2}{2}) dy \\ & + \beta \int_{\Omega} (D^{\nu}I - \frac{d_1+d_2}{2}) dy + \gamma g \cdot \operatorname{div}\left(\frac{\nabla\phi}{|\nabla\phi|}\right)] \\ & + \mu[\Delta\phi - \operatorname{div}\left(\frac{\nabla\phi}{|\nabla\phi|}\right)]. \end{aligned} \quad (40)$$

where d_1 and d_2 are from Eq. (15). The results of our model with fractional differentiation based fitting E^{Fr} and E^{FF} are shown in Section V.

IV. NUMERICAL ALGORITHM

To avoid singularities, we modify $|\nabla\phi|$ with a small value $s = 10^{-3}$ and it is written as

$$|\nabla\phi| = \sqrt{(\nabla_x\phi)^2 + (\nabla_y\phi)^2 + s}. \quad (41)$$

All the spatial partial derivatives are approximated by the central difference, for example,

$$\begin{aligned} \nabla_x\phi(i, j) &= \frac{\phi(i+1, j) - \phi(i-1, j)}{2} \\ \nabla_y\phi(i, j) &= \frac{\phi(i, j+1) - \phi(i, j-1)}{2} \end{aligned} \quad (42)$$

$D^{\nu}I$ is calculated using fractional calculus [35]–[38].

Three commonly used definitions for fractional calculus are Grunwald-Letnikov (G-L), Riemann-Liouville (R-L) and Capotu respectively [39]. Among them, the most frequently used in image processing is the G-L definition, formulated as

$${}_G f_t^{\nu} = \lim_{h \rightarrow 0} h^{\nu} \sum_{j=0}^{\frac{t-a}{h}} \frac{\Gamma(\nu+j)}{j!\Gamma(\nu)} f(t-jh), \nu \in \mathcal{R}^-. \quad (43)$$

In the two-dimensional image domain, the ν -order spatial partial derivatives in directions of x and y are approximated by

$$\begin{aligned} D_x^{\nu}I(x, y) \approx & I(x, y) + (-\nu)I(x-1, y) \\ & + \frac{(-\nu)(-\nu+1)}{2}I(x-2, y) + \dots \\ & + \frac{\Gamma(-\nu+1)}{n!\Gamma(-\nu+n+1)}I(x-n, y), \end{aligned} \quad (44)$$

$$\begin{aligned} D_y^{\nu}I(x, y) \approx & I(x, y) + (-\nu)I(x, y-1) \\ & + \frac{(-\nu)(-\nu+1)}{2}I(x, y-2) + \dots \\ & + \frac{\Gamma(-\nu+1)}{n!\Gamma(-\nu+n+1)}I(x, y-n). \end{aligned} \quad (45)$$

0	$\frac{\nu^2 - \nu}{2}$	0
0	$-\nu$	0
0	1	0

(a) Mask of x-direction

0	0	0
$\frac{\nu^2 - \nu}{2}$	$-\nu$	1
0	0	0

(b) Mask of y-direction

Fig. 1. Masks of fractional order in x and y directions.

Algorithm 1 The Gradient Descent Algorithm for Solving the Proposed Model

Input: $\phi_0, c_0, \sigma, \nu, \lambda, \beta, \gamma, \mu, iter_{max}, \tau, tol$
Output: ϕ
 Compute $D^\nu I$ by Eq. (46)
for $k = 1 : iter_{max}$ **do**
 Update f_1 and f_2 by Eq. (27);
 d_1 and d_2 by Eq. (28);
 ϕ_k by Eq. (47)
 err by Eq. (48)
 If $err \leq tol$
 break
end
end

The ν -order derivative and its amplitude are given as follow

$$D^\nu I = [D_x^\nu I, D_y^\nu I]^\top, \quad |D^\nu I| = \sqrt{D_x^\nu I^2 + D_y^\nu I^2}. \quad (46)$$

Here we take the amplitude as the value of $D^\nu I$.

Considering the amount of computation, we take the first three terms for Eq. (44) and Eq. (45). This is done because if the ν -order derivative of each pixel on the image domain is calculated as Eq. (44) and Eq. (45), there are n pixels involved besides the coefficients before every term, which requires a large amount of computation. Pixels far away have smaller effects on the current pixel relatively than the first three neighbors.

Now we construct two 3×3 fractional order differentiation masks in x and y directions in Fig. 1. In order to solve the proposed model, the widely used gradient descent algorithm [40] is applied. The right sides of Eq. (39) and Eq. (40) are denoted as L_1 and L_2 respectively, which are computed in Case 1 and Case 2 separately. The two cases are referred to L for calculation. The $(k+1)$ -th iteration of ϕ is computed by

$$\phi^{k+1} = \phi^k + \tau L \quad (47)$$

where $\tau > 0$ is the step length. It is very important that the iteration stops within a finite number of steps. Denoting the maximum of iterations as $iter_{max}$, the termination criterion is defined as follows

$$err = \frac{\|\phi^{k+1} - \phi^k\|_F}{\|\phi^k\|_F} \quad (48)$$

i.e., the iteration stops when $err \leq tol$ and the number of iterations is no more than $iter_{max}$, where tol is the tolerance. The iterative steps are organized in Algorithm 1.

V. EXPERIMENTS

In this section, we present the results of the proposed model and compared models for segmentation when images are clean

and when images include Gaussian noise, speckled noise as well as salt and pepper noise. All the numerical experiments were performed in MATLAB (R2016a) on a windows7 (64bit) desktop computer with an Intel Core i5 3.20 GHz processor and 4.0GB of RAM. The compared CV, LRCV, [29] and [30] models are programmed by ourselves. The test images in Fig. 2 and their ground truths are from [42] (pic1 - pic3 and pic6 - pic14) and [43] (pic4 - pic5). The initial curves are shown with red boxes in them. Variances of noise are set as 0.01 by MATLAB command *imnoise*. Parameters of our model for experiments are default to $\lambda = 25$, $\beta = 1$, $\gamma = 0.0002 \times 255^2$, $\mu = 0.1$ and $\tau = 0.02$. σ in K_σ is tunable.

As for the evaluation of results, the dice similarity coefficient (DSC) [44] metric and computational time are provided to assess the accuracy and efficiency, respectively. Supposing that A and B are segmented result and ground truth respectively, the DSC is computed by

$$DSC = \frac{2|A \cap B|}{|A| + |B|} \quad (49)$$

where $|\cdot|$ is the number of pixels. The range of the DSC is $[0, 1]$. It is easy to find that the larger the DSC is, the more accurate the segmentation.

A. Results of the Proposed Model

In this subsection, the experiments of the proposed model are performed. In order to show the effectiveness of the Gaussian kernel K_σ in E^{Fr} , which is discussed in Case 1 and Case 2 of Remark 2, comparison experiments are conducted for the two cases.

In Fig. 3 - Fig. 5, the four columns from left to right are results of clean images, and images with Gaussian noise, speckled noise and salt and pepper noise, respectively. The first row is results of Case 1 and the second row is results of Case 2. As seen in Fig. 3 - Fig. 5, results of Case 2 for noisy images are affected by noise to varying degrees and some noise is mistaken as edges of images. These results show that the Gaussian kernel K_σ in E^{Fr} is effective at helping curves to deal with noise during evolution.

In addition to visual effects, the DSC and computational time are compared for the two cases in Table I. It is seen that Case 1 results in larger DSCs. Case 1 takes less time for clean images. This means that K_σ in E^{Fr} can accelerate the evolution of curves. The two cases achieve similar computational time for noisy images. There are two reasons for this. First, during the iterations, Gaussian kernel K_σ in Case 1 is additionally calculated compared with Case 2. Second, curves may stop before reaching right edges in Case 2 for noisy images.

From experiments of Case 1 and Case 2, it is seen that Case 1 obtains better segmentations. It can be concluded that the Gaussian kernel K_σ can accelerate the evolution of curves and help to obtain more precise segmented results. This demonstrates the significance of K_σ in E^{Fr} .

The proposed model is also capable of segmenting color images with different types of noise. In Fig. 6, results of color images pic4 and pic5 are shown. The four columns are results

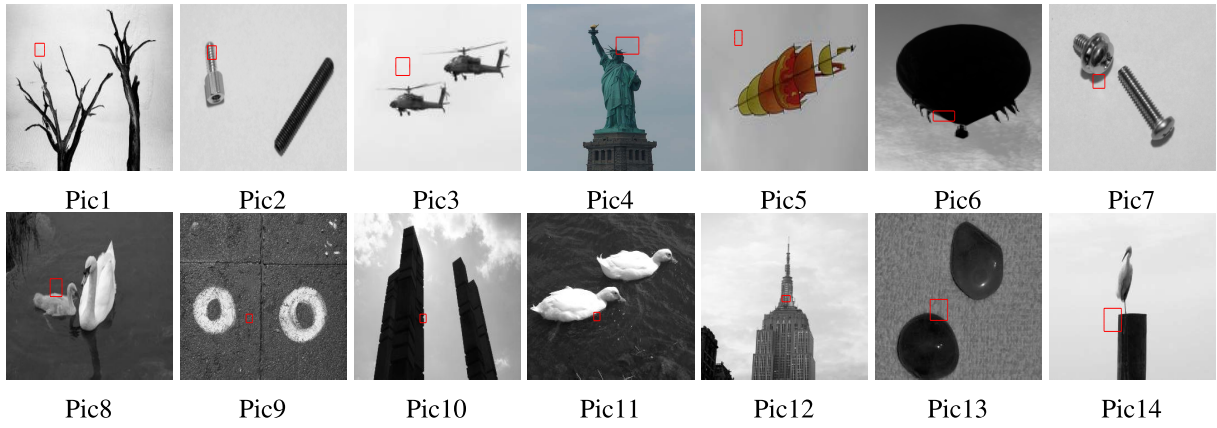


Fig. 2. Test images with initial positions of contours.

TABLE I
COMPARISON OF DSC AND COMPUTATIONAL TIME (SECONDS) FOR CASE 1 AND CASE 2

Images	Cases	DSC				Time			
		Clean	Gaussian	Speckle	Salt	Clean	Gaussian	Speckle	Salt
Pic1	Case 1	0.9357	0.9277	0.9360	0.9209	1.0802	1.2301	1.1655	1.2158
	Case 2	0.9368	0.9257	0.9354	0.9206	1.1317	1.2149	1.1674	1.2062
Pic2	Case 1	0.9349	0.8810	0.8822	0.8520	0.8565	0.9396	0.9397	0.9830
	Case 2	0.9340	0.8695	0.8783	0.8295	0.8708	0.9811	0.9760	0.9942
Pic3	Case 1	0.9218	0.9162	0.9153	0.9148	1.1197	1.2446	1.1609	1.2319
	Case 2	0.9202	0.9118	0.9131	0.8995	1.1504	1.2219	1.1720	1.2177

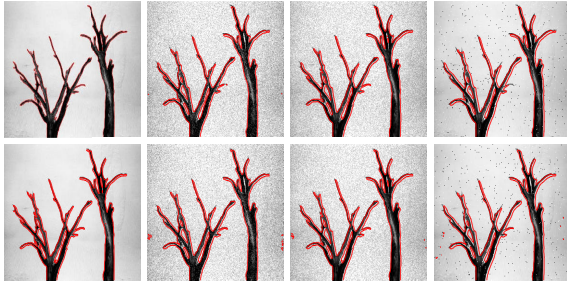


Fig. 3. Results of Case 1 and Case 2 for Pic1.

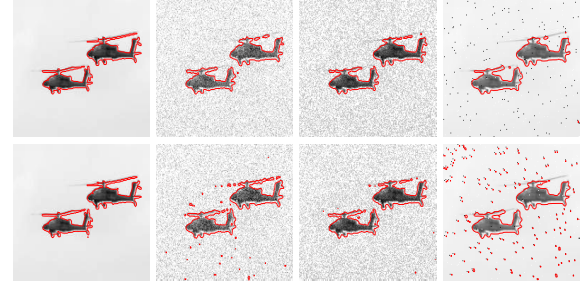


Fig. 5. Results of Case 1 and Case 2 for Pic3.

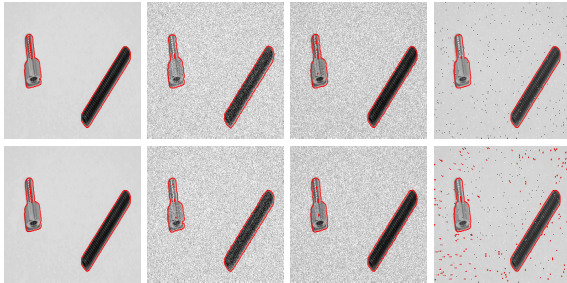


Fig. 4. Results of Case 1 and Case 2 for Pic2.

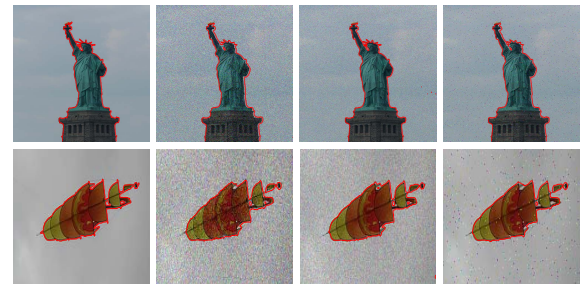


Fig. 6. Results of the proposed model for Pic4 and Pic5.

of clean images, images with Gaussian noise, speckled noise and salt and pepper noise, respectively.

B. Results of the Compared Models

In this subsection, we compare the proposed model (Case 1) with CV, LRCV, [29] and [30] with different types of noise.

In Fig. 7 - Fig. 10, the five rows from top to bottom are segmented results of CV, LRCV, [29], [30] and the proposed model, respectively. In Fig. 7, the results for clean images show that the proposed model obtains more complete contours. Fig. 8 - Fig. 10 are segmented results of images with

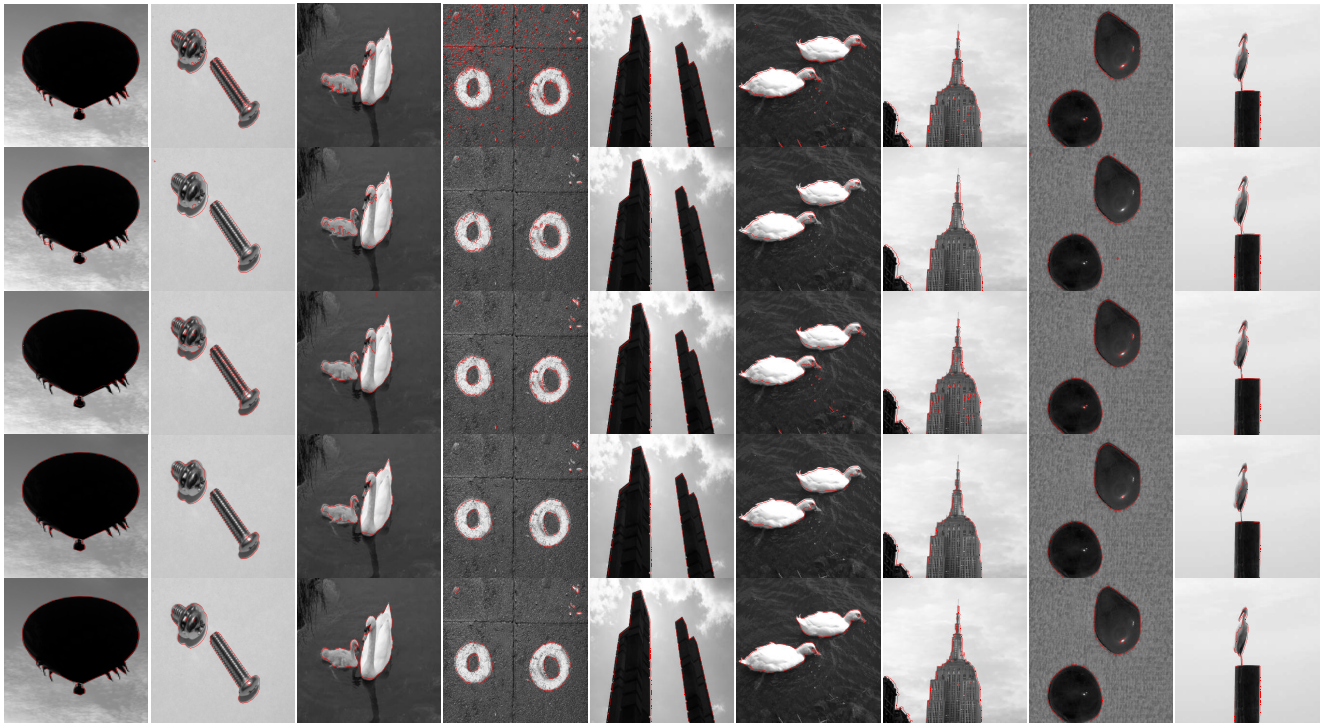


Fig. 7. Compared results for clean images.

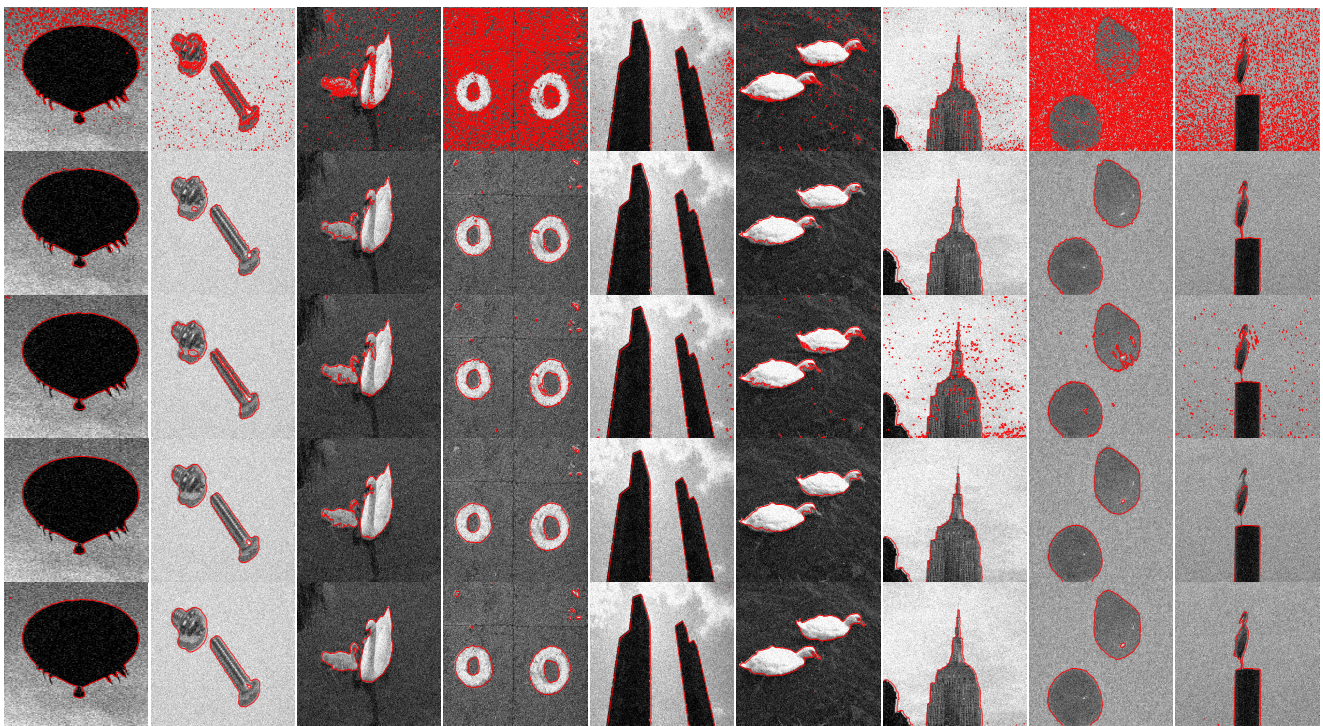


Fig. 8. Compared results for images with gaussian noise.

Gaussian, speckled and salt and pepper noise, respectively. Among them, results of CV, LRCV and [29] are susceptible to noise interference, while [30] and the proposed model perform better.

In addition to the visual effects, the DSC and computational time are compared for clean and noisy images in Table II.

It is seen that the DSCs of our model are almost the largest and computational time is almost the shortest among these models. The CV and LRCV models take less computational time, however, the segmentations shown in Fig. 7 - Fig. 10 are not always desired in case of images with noises. While [29] and [30] take more computational time than our model.

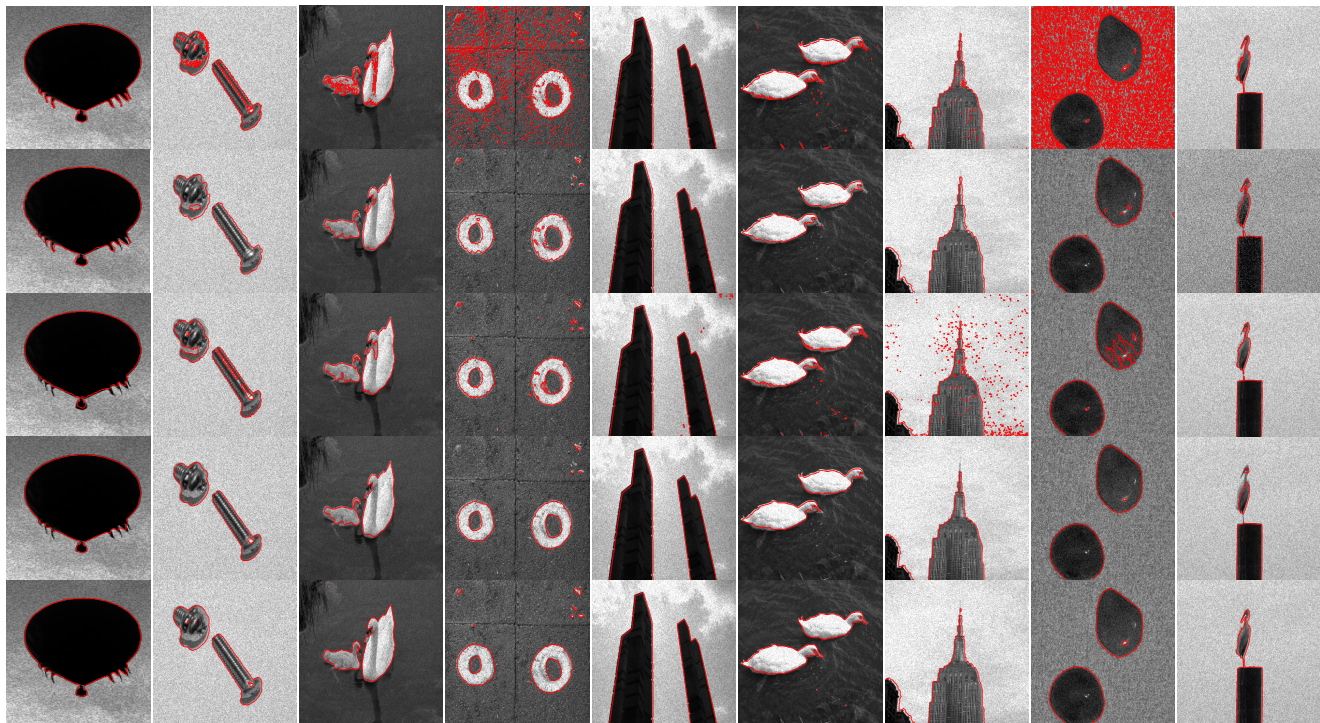


Fig. 9. Compared results for images with speckle noise.

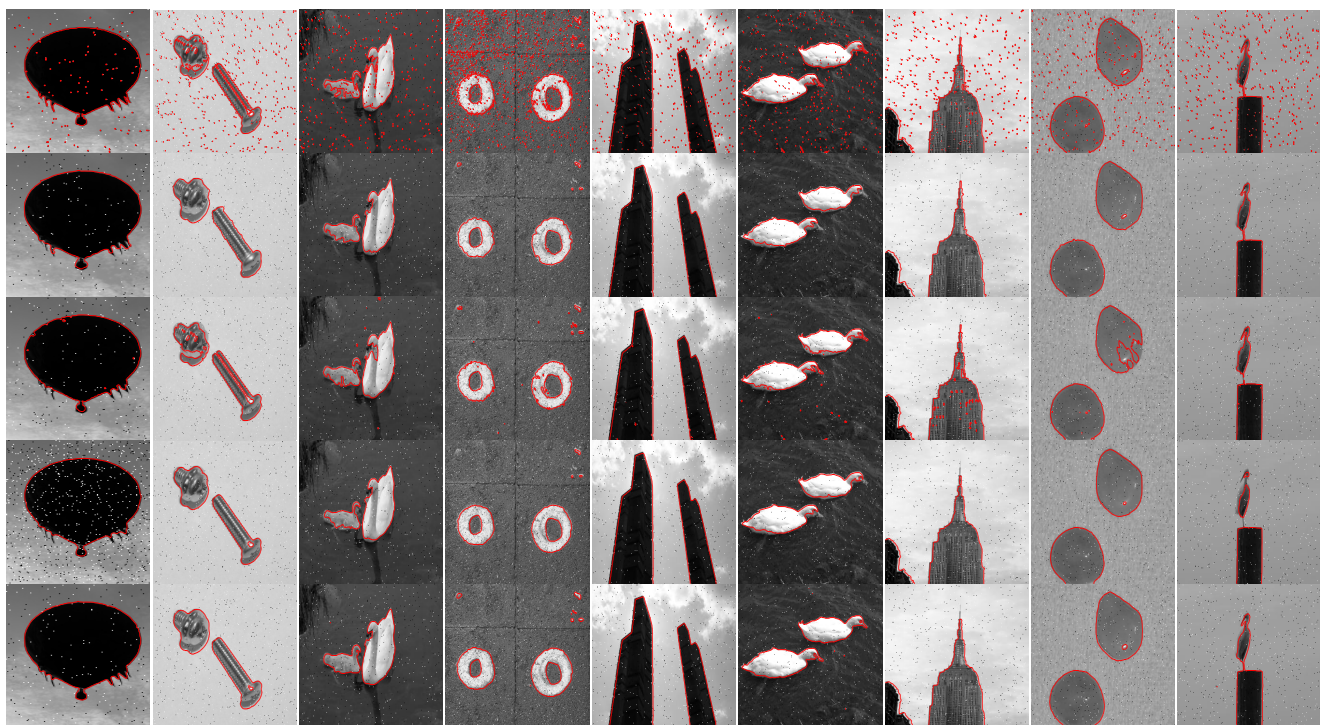


Fig. 10. Compared results for images with salt and pepper noise.

The differences in segmented effect evaluated by the DSC and computational time among the compared models are attributed to the formulation of modeling. Different from the CV and LRCV models, fractional order differentiation is incorporated into the proposed model, which makes it effective at segmenting images with different types of noise. Instead

of selecting a fixed constant as the fractional differentiation order, a simple definition of fractional order is designed to adaptively process pixels inside the image domain, which saves significant time for experiments. Furthermore, in [30], a Gaussian kernel is used to regularize the level set function after each iteration, which is avoided by introducing a penalty

TABLE II
COMPARISON OF DSC AND COMPUTATIONAL TIME (SECONDS)

Image	Models	DSC				Time			
		Clean	Gaussian	Speckle	Salt	Clean	Gaussian	Speckle	Salt
Pic6	CV	0.9880	0.9638	0.9867	0.9795	0.8032	0.8378	0.8148	0.8147
	LRCV	0.9898	0.9895	0.9909	0.9923	0.9488	0.9734	0.9733	0.9764
	[29]	0.9782	0.9815	0.9803	0.9795	1.5806	1.5817	1.5871	1.5666
	[30]	0.9905	0.9851	0.9842	0.9847	1.0109	2.2417	1.8935	2.9624
	Ours	0.9918	0.9900	0.9910	0.9912	0.6318	0.7007	0.7003	0.6716
Pic7	CV	0.9377	0.8074	0.9164	0.8856	1.0337	1.1613	1.0275	1.0420
	LRCV	0.9682	0.9573	0.9525	0.9732	0.9676	1.3863	1.5270	1.5636
	[29]	0.8719	0.8800	0.8780	0.8650	3.3232	3.2810	3.2466	3.2600
	[30]	0.9433	0.9468	0.9448	0.9436	40.8582	42.0231	39.8184	40.5997
	Ours	0.9867	0.9833	0.9836	0.9842	1.0015	1.0383	1.0541	1.0935
Pic8	CV	0.9350	0.8441	0.8472	0.8467	0.9551	1.0147	1.0937	1.0757
	LRCV	0.7002	0.8088	0.8447	0.8295	1.1931	1.2033	1.2124	1.2117
	[29]	0.9281	0.9128	0.9101	0.9141	2.5064	2.5198	2.5270	2.5388
	[30]	0.9227	0.9200	0.9219	0.9235	23.7575	18.3115	19.3293	21.4048
	Ours	0.9542	0.9510	0.9551	0.9540	0.8661	0.9324	0.8978	0.9036
Pic9	CV	0.8439	0.4799	0.7775	0.8467	1.0466	0.9968	1.0461	1.0602
	LRCV	0.8673	0.9247	0.8922	0.9143	1.2233	1.2634	1.2418	1.2429
	[29]	0.9070	0.9054	0.9053	0.9010	2.6155	2.6346	2.6394	2.6014
	[30]	0.9392	0.9404	0.9397	0.9393	10.9012	35.3099	29.6295	31.5238
	Ours	0.9533	0.9500	0.9529	0.9539	0.9064	0.9681	0.9627	0.9752
Pic10	CV	0.9997	0.9865	0.9988	0.9908	0.9189	0.9768	0.9517	1.0035
	LRCV	0.9863	0.9886	0.9879	0.9932	1.2251	1.2741	1.2567	1.2490
	[29]	0.9984	0.9934	0.9969	0.9979	1.7179	2.6199	2.6220	2.6338
	[30]	0.9968	0.9967	0.9984	0.9972	3.8722	3.8615	9.9483	3.6510
	Ours	0.9988	0.9955	0.9955	0.9943	0.7760	0.8461	0.8563	0.8416
Pic11	CV	0.9431	0.9167	0.9395	0.9193	0.9029	0.9029	0.9851	0.9407
	LRCV	0.9176	0.9240	0.9345	0.9264	1.2253	1.2553	1.2545	1.2465
	[29]	0.9189	0.8920	0.9176	0.8948	2.6439	2.6913	2.6577	2.66118
	[30]	0.9172	0.9222	0.9257	0.9153	11.2230	11.8105	10.3440	10.7173
	Ours	0.9585	0.9600	0.9601	0.9583	0.8869	0.9025	0.9101	0.9200
Pic12	CV	0.9873	0.9608	0.9838	0.9744	0.8784	0.9196	0.9091	0.9011
	LRCV	0.9692	0.9755	0.9720	0.9736	1.2269	1.2431	1.2365	1.2593
	[29]	0.9807	0.9416	0.9534	0.9798	2.5904	2.6196	2.6394	2.6027
	[30]	0.9893	0.9876	0.9890	0.9875	6.3761	9.3653	4.5189	11.0671
	Ours	0.9872	0.9820	0.9821	0.9781	0.8268	0.8979	0.9097	0.9327
Pic13	CV	0.9656	0.6246	0.6923	0.9554	1.0532	1.0234	1.0666	1.0620
	LRCV	0.9797	0.9688	0.9747	0.9687	1.2667	1.3972	1.4102	1.4117
	[29]	0.9614	0.9493	0.9241	0.9253	2.7877	2.8699	2.8307	2.8144
	[30]	0.9650	0.9633	0.9662	0.9645	39.4847	40.7135	39.5184	42.2512
	Ours	0.9776	0.9764	0.9771	0.9776	1.1671	1.2007	1.2150	1.1993
Pic14	CV	0.9665	0.6637	0.9650	0.9383	0.9180	0.9600	0.9412	0.9532
	LRCV	0.9637	0.9609	0.9500	0.9605	1.2432	1.2736	1.2782	1.2833
	[29]	0.9725	0.9400	0.9672	0.9717	2.1272	2.7206	2.6724	2.7017
	[30]	0.9391	0.9300	0.9374	0.9386	36.1722	36.0935	33.4585	34.8083
	Ours	0.9790	0.9651	0.9745	0.9669	0.8831	0.998	0.9668	1.0015

term in [29] and the proposed model. The gradient flow of the proposed model is simplified to improve the efficiency as well. Therefore the proposed model obtains larger DSCs and takes less computational time.

VI. CONCLUSION

In this paper, a novel active model is proposed for noisy image segmentation. There are three main points to be summarized. First, the fractional differentiation with an adaptively defined order is incorporated into the fitting term to deal with noise during the evolution of curves. The adaptability avoids repeated experiments to select a suitable fractional order, which is only computed once according to its definition. Second, the Gaussian kernel in the fractional differentiation based fitting E^{Fr} makes the curves evolve towards the right edges of images faster with less affects of noise. Finally, experiments of comparing the proposed model with the CV model, the LRCV model, the fractional ACMs [29] and [30] demonstrate the effectiveness and efficiency of the proposed model. In our future work, modeling and adaptive weighting techniques will be explored to achieve faster calculations and use fewer parameters.

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Meng-Meng Li received the B.S. degree from the Department of Mathematics, Qufu Normal University, Jining, China, in 2015. She is currently pursuing the Ph.D. degree with the School of Mathematics and Statistics, Beijing Institute of Technology, Beijing, China. Her research interests include image processing, fractional calculus, and partial differential equation.



Bing-Zhao Li (Member, IEEE) was born in 1975. He received the B.S. degree from Shandong Normal University in 1998, and the M.S. and Ph.D. degrees from the Beijing Institute of Technology, in 2001 and 2007, respectively. He is currently a Professor with the School of Mathematics and Statistics, Beijing Institute of Technology, Beijing, China. His current research interests include the theory and applications of fractional Fourier transform, linear canonical transform, and fractional calculus.